

# Multi-modal Egocentric Action Recognition

Antonio Dal Maso  
Politecnico di Torino

s311444@studenti.polito.it

Gabriele Cirotto  
Politecnico di Torino

s307732@studenti.polito.it

Teresa Argnani  
Politecnico di Torino

s313701@studenti.polito.it

## Abstract

*Egocentric action recognition involves understanding and categorizing human actions from a first-person perspective video, and often comes with many related challenges such as complex backgrounds, shaky camera movements and hand-object interactions.*

*In this paper, we aim to address this problem by leveraging different data sources, by working with the standard visual modality of RGB streams and investigating the ElectroMyoGraphic signals (EMG) representing forearm muscle activity, which is receiving more attention in the computer vision community nowadays. Firstly we consider those modalities individually and then we combine them together to exploit their complementary information in a multimodal fashion.*

*We run our experiments using two egocentric video datasets: EpicKitchens and ActionSense (which is the one including the EMG measurements).*

*Our code is available at <https://github.com/antoniodalmaso/mldl23-ego>.*

## 1. Introduction

Nowadays, egocentric video recordings have gained popularity in the computer vision community due to their utility in various areas such as health monitoring [19], extreme sport performance analysis [16], life logging [17], and home automation [15]. For this reason, it is interesting to develop new methods or adapt existing ones to the first-person point of view scenario. Moreover, the video domain is naturally appropriate for building models that incorporate different modalities, since videos encompass visual information, such as appearance and motion, as well as audio information, which increases the complexity and dimensionality of the problem.

In the first part of this work, our focus is on a single visual modality: we consider videos as ordered collections of RGB frames. To conduct our experiments, we utilize EpicKitchens-55 [4], a large-scale egocentric video dataset that captures daily kitchen activities. In particular, we focus on

its 8 most common actions and on a subset of the dataset corresponding to three of its domains (D1, D2 and D3 from the smallest to the largest), as described in MM-SADA [14]. We take advantage of the pre-trained I3D architecture proposed in [3] to construct compressed representations for each sample and employ the obtained features to train the final classifier.

In the second part, we shift our analysis to a different modality and explore the ActionSense dataset [6]. This dataset also provides first-person video recordings in a kitchen environment, along with synchronized measures obtained from a diverse set of sensors worn by the actors, capturing motion, force, and attention information. Specifically, we focus on the ElectroMyoGraphic (EMG) signals, which are obtained from a Myo Gesture Control Armband by Thalmic Labs [13], worn on each forearm by the person performing the actions. EMG signals have various potential applications, including controlling prosthetic limbs or assistive devices, assessing muscle activation patterns during rehabilitation therapy, and detecting early signs of muscle-related disorders. Consequently, the study of EMG signals has gained popularity in recent times.

Initially, we train a classification model that takes only the EMG modality as input by computing a 2D CNN on the spectrograms of the signals. Finally, we build a multi-modal classifier for ActionSense that combines both RGB and EMG proposed architectures, synchronizing the inputs.

In the following sections, we discuss the implementation details as well as the achieved results.

## 2. Related work

**Video classification.** Current video architectures are founded mainly on two approaches, depending on whether the convolutional and layers operators use 2D (image-based) or 3D (video-based) kernels [12].

In the case of 2D convolutional methods, it is also crucial to incorporate the temporal information propagated across frames, which can be done either using temporally-recurrent layers such as Long Short-Term Memories (LSTMs) [1], or feature aggregation over time.

Regarding the 3D approach, we mention the Two-Stream

Inflated 3D ConvNets (I3D) model [3], which is an example of an architecture that extends 2D CNN by inflating the filters in order to perform 3D convolutions, capturing both spatial and temporal features simultaneously. This model utilizes a two stream architecture: a branch for processing RGB streams and one for processing optical flow (that captures motion information); in our proposal we take advantage of this model, but we ignore the flow modality because it is not among the ones we want to explore. The I3D model is often initialized with weights pretrained on large-scale image recognition tasks, such as the Imagenet dataset [7].

**First Person Action Recognition.** The complex nature of egocentric videos raises a variety of challenges, such as complex backgrounds, shaky camera movements and hand-object interaction which limit the performance of traditional, third-person, approaches. While the use of optical flow has been the standard procedure in first person action recognition [11], the interest has recently shifted towards more lightweight alternatives to be combined with the common RGB modalities, such as gaze, audio [10] or depth [8] to enable motion modeling in online settings.

**Action Recognition through EMG signals.** Human action recognition has been an active field of study in machine learning, but still limited work has been done for recognizing actions in performing tasks that involve complex, intricate operations, especially in a manufacturing environment. The authors of [20] and [2] proposed different approaches for action recognition based on users' data captured by various sensors. Specifically, they combined both EMG signals, Inertial Measurement Unit (IMU), and skeleton data.

**Multimodal architectures.** The Temporal Binding Network (TBN) [10] focuses on multimodal fusion for egocentric action recognition and explores combining audio and visual inputs within a Temporal Binding Window (TBW), allowing the model to train using asynchronous inputs from the various modalities. TBN produces a single prediction per window after fusing all modality representations, where the audio stream is represented through its spectrogram, and predictions from multiple windows, possibly overlapping, are averaged.

### 3. Proposed method

#### 3.1. RGB features

RGB data is surely the main actor when we talk about First Person Action Recognition and computer vision in general, and it is indeed our starting point. We used I3D to process the RGB data and to extract its features for both Epic-Kitchens-55 and ActionSense.

The process of extraction is the following:

- Given a video, it is divided into 5 segments of equal length, named clips.

- Since processing long sequences of frames can be time and resource-intensive, it is crucial to conceive an appropriate strategy to meaningfully sample a small subset of them; in our case, 16 frames are selected from each clip using dense sampling (the frames are adjacent and eventually separated by a small stride), which focuses on the appearance of the video as the frames are close in time.
- Each set of frames is fed into I3D, which produces a vector of 1024 elements as the output of its last convolutional layer.
- The resulting  $5 \times 1024$  values are then used to represent the video.

Even though such features are not easily interpretable being the output of a neural network, we can still try to understand if they constitute a reasonable representation of the starting dataset. In order to do so, we have reduced the dimensionality of the samples using PCA (maintaining 90% of the original explained variance), and then we have applied the KMeans algorithm to look for clusters. Figure 1 represents the distributions of the class labels inside each cluster, and we can observe that each group can be associated almost exclusively with one label.

Moreover, Figure 2 shows the representation of each video in the space of the first two principal components, where each video is depicted using its central frame. We can appreciate how frames that are close to each other actually have similarities in their aspects, again justifying the validity of the feature representation.

The results discussed above are related to the train split of D1 (Epic-Kitchens-55), but similar considerations can be done for the other splits too.

#### 3.2. RGB classifier

Given our limited resources both in terms of time and computational power, we could not develop a neural network including I3D in its architecture to perform the task; thus, we limited to train a classifier which takes as input the features extracted in the previous step. We put to the test two architectures differing in the strategy used to combine the features coming from the different clips of a video:

- **Concatenation;** the 5 vectors are simply concatenated into a single one before entering the fully connected layers implementing the classifier.
- **LSTM;** the first layer of the previous solution is substituted with a recurrent layer to merge the 5 vectors. This is reasonable because, even though the clips composing a video may overlap, they are still sorted in chronological order.

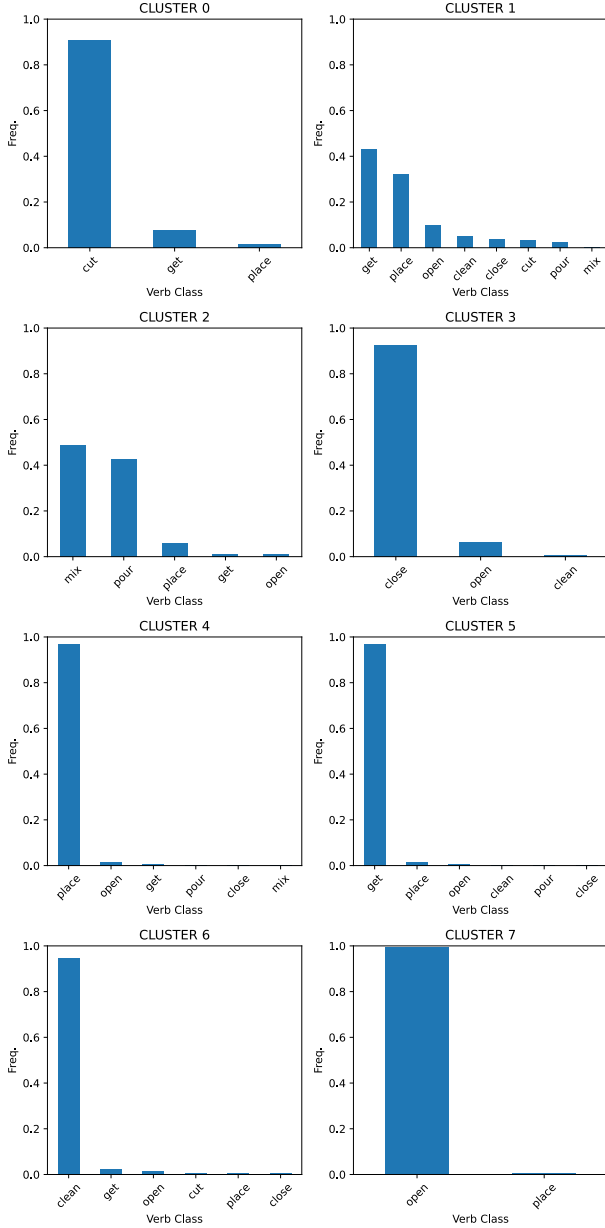


Figure 1. Distributions of the class labels inside the clusters found for the train split of D1 using KMeans algorithm.

### 3.3. EMG features

Among the other modalities, ActionSense provides 16 streams of electromyographic data, 8 for each forearm, consisting in measures representing the electrical activity produced by muscles.

In order to use this data, we have applied the same preprocessing pipeline proposed in [6]: each stream is rectified by taking the absolute value, and then a low-pass filter with cutoff frequency of 5 Hz is applied; all 8 streams from an

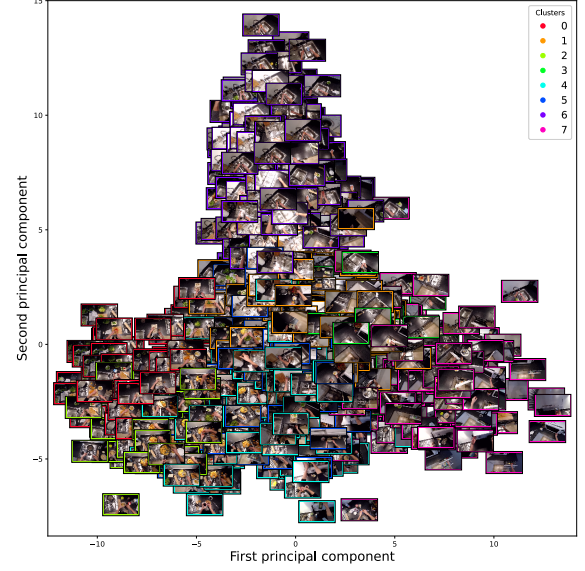


Figure 2. Representation of the actions of the train split from the D1 domain. Each action is depicted using the central frame of the corresponding video, and the axes represent the first two principal components. The colors assigned to the data points are based on predictions made by the KMeans algorithm (the numbers of the clusters are consistent with the ones in Figure 1).

armband are then jointly normalized and shifted to the range  $[-1, 1]$  using the minimum and maximum values across all streams. Next, all the streams are resampled to a frequency of  $\approx 150$  Hz (5 times the one of RGB data). Notice that keeping an high sample frequency is important because, in the next steps of our analysis, spectrograms (Figure 3) will be computed upon these streams, and we want them to be large and rich of information.

The result of the preprocessing is that to each sample corresponds a matrix of  $N \times M$  values,  $N$  being the number of measures collected in the timespan covered by the action and  $M = 16$  being the number of streams.

### 3.4. EMG classifier

We put to the test 2 architectures to exploit this modality:

- **LSTM**; the samples of the dataset are considered as series of measures, each one composed of 16 values (one per sensor). These vectors are combined together using a LSTM, and the result is fed into a fully connected layer which computes the predictions.
- **2D CNN**; it can be observed that each of the EMG streams can be considered as a time series, and as such it can be processed in a similar way to audio recordings. In particular, we compute the spectrograms of

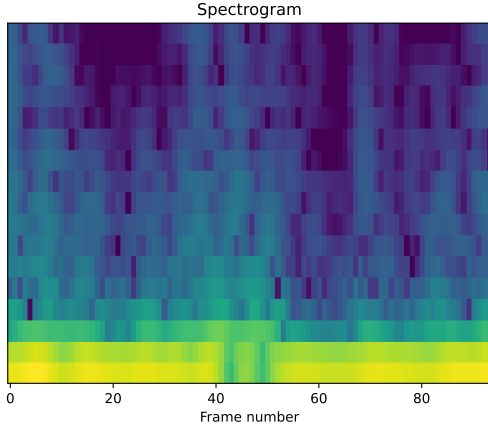


Figure 3. Visualization of a single spectrogram representing the signal strength over time (30 frames are  $\simeq 1$  second) at various frequencies.

the streams, and then we use a 2D convolutional neural network to process them (similar to what [10] does with audio). We have actually identified 2 ways of manipulating the EMG signals to turn them into spectrograms:

- The streams corresponding to each forearm can be summed together to obtain a measure of the overall activation; the result consists in 2 time series, on top of which we can compute the spectrograms.
- The streams can be directly used to compute the spectrograms.

In practice, the difference between the two methods is the number of channels of the resulting “images” (2 and 16, respectively).

### 3.5. Multimodal classifier

Finally, we propose a multimodal classifier using both the RGB and the EMG samples of ActionSense, combining the architectures introduced in the previous sections that achieved the best results in terms of accuracy. In particular:

- The RGB data are processed by I3D in the same way we did for Epic-Kitchens; the 5 vectors associated with each sample are simply concatenated together.
- The EMG samples are converted into 16-channels images by computing their spectrograms, and are then processed by the same 2D CNN introduced in Section 3.4.
- The features obtained in the 2 previous steps are concatenated together and fed to a series of fully connected layers implementing the actual classifier.

The resulting architecture is very similar to what has been done by the authors of [10] for TBN.

## 4. Experimental results

This section discusses first of datasets and architecture in Section 4.1, then the results are presented in Section 4.2.

### 4.1. Implementation Details

**Datasets.** Regarding the Epic-Kitchens-55 dataset, the three domains we used are characterized by the cardinalities shown in Table 1.

| Split | # Samples in train split | # Samples in test split |
|-------|--------------------------|-------------------------|
| D1    | 1543                     | 435                     |
| D2    | 2495                     | 750                     |
| D3    | 3897                     | 974                     |

Table 1. Cardinality of considered Epic-Kitchens-55 splits.

However, the central point of our analysis is the multi-modal domain, thus we are now going to look at the details of the version of the ActionSense dataset we used for our experiments: we focused on the split S04, which is characterized by only 59 samples belonging to 20 class labels. In order to obtain a number of samples suitable to train a model, we first divided each record in 20 possibly overlapping segments of 2.5 seconds (which is slightly less than the duration of the shortest action in the split); next, we selected 20 segments for every class label among all the related available ones, in order to have a balanced class distribution (Figure 4). We also included an additional class, namely the *None* label, corresponding to the measures that are not associated to any action. Also for this last class we collected 20 short segments.

The resulting dataset is composed of 420 samples, each of which is a  $374 \times 16$  matrix.

**RGB Architecture.** As discussed in Section 3.1, we used I3D to extract the RGB features for Epic-Kitchens-55 and ActionSense. In particular, such network has BNInception as backbone and was pre-trained on Kinetics and Imagenet, two datasets well-known in the literature for their performances in the domains of temporal context and image classification, respectively. Our experiments consist in maintaining the original architecture of I3D described in [3] up until the last average pooling layer, and then developing and testing different prediction layers. Our proposals are:

- A simple sequence of four fully connected layers, of which the first three are characterized by a dropout rate of 0.5 and the ReLU activation function. After the last fully connected layer, the logits are passed to a log softmax activation function which has been demonstrated to work well with high dimensional features [5].

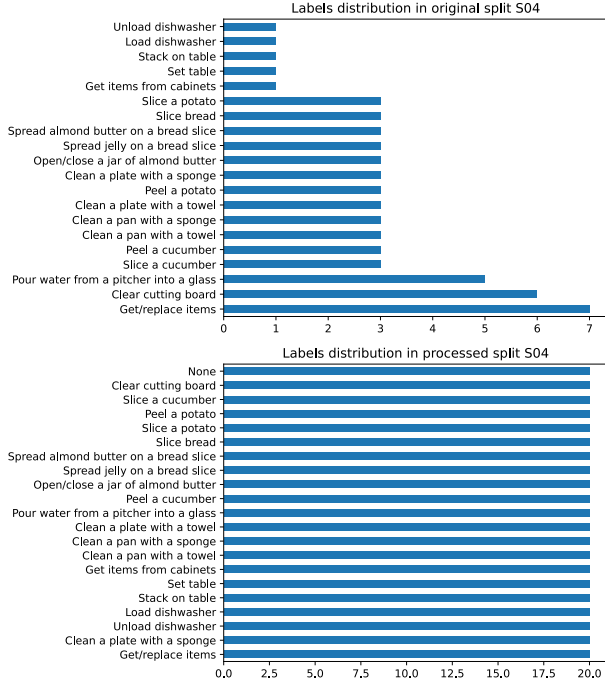


Figure 4. Distribution of the actions’ label in the split S04, before and after the processing proposed to make it balanced.

- The second approach tries to exploit the temporal context of the extracted features passing them to a LSTM model before applying two fully connected layers and the log softmax.

All the results for both Epic-Kitchens-55 and ActionSense are shown in Table 2, where different experiments varying the number of epochs have been conducted. In general, we can notice that the LSTM architecture is slightly underperforming with respect to the simple linear architecture in D1 and ActionSense, while after some epochs it is better for both D2 and D3. It is possible that since I3D already extracts useful temporal context, the LSTM does not produce significant improvements.

**EMG Architecture.** We propose two main approaches:

- The first architecture we implement for the classification of the EMG signals is a naïve approach similar to the one proposed in [6], exploiting a simple LSTM that turns each  $374 \times 16$  sample into a one-dimensional array of dimension 50. Next, a single fully connected layer turns such array into the 21 class logits for the final softmax function.

All the experiments regarding this network were run by using the cross entropy loss and Adam as optimizer (learning rate  $lr = 0.01$ ).

- The second proposed network takes in input the EMG

| Model      | Epochs | D1           | D2           | D3           | S04          |
|------------|--------|--------------|--------------|--------------|--------------|
| I3D-Linear | 10     | <b>57.70</b> | 68.80        | 72.69        | 27.38        |
| I3D-Linear | 20     | 57.01        | 68.53        | 72.28        | 59.52        |
| I3D-Linear | 40     | 57.47        | 68.53        | 72.79        | 73.81        |
| I3D-Linear | 80     | -            | -            | -            | <b>79.76</b> |
| I3D-LSTM   | 10     | 56.32        | 65.60        | 72.69        | 10.71        |
| I3D-LSTM   | 20     | 56.10        | 69.20        | 71.15        | 57.14        |
| I3D-LSTM   | 40     | 55.63        | <b>69.47</b> | <b>72.89</b> | 73.81        |
| I3D-LSTM   | 80     | -            | -            | -            | 71.43        |

Table 2. Comparison of the two approaches adopted for the RGB stream. We can notice that the LSTM reaches slightly better results in EK-55 dataset, while for the split S04 the linear prediction layer outperforms the LSTM network for any number of epochs. All the floating point values indicate mean accuracy on the test set.

spectrograms, and its architecture is described in Table 4; we refer to it as SpecNet and it is a 2D CNN that interprets the spectrograms as images with 16 channels. Following the approach of MobileNetV3 [9], we use as activation function the hard version of the swish non-linearity, namely the *hardswish*:

$$\text{h-swish}(x) = x \frac{\text{ReLU6}(x + 3)}{6}$$

The implementation of this non-linearity helps the architecture to improve the accuracy while maintaining a low computational cost. Another important point is the presence of the batch normalization after the first and the last convolutional layers, which has been used in other works to improve the overall accuracy of shallow networks while increasing the speed at which they train [18].

As anticipated in Section 3.4, we also tested a version of SpecNet working on images having 2 channels. We refer to it as SpecNet2Ch, and it differs from the original network only in the dimensions of the input of the fully connected layers.

The experiments for the just described architectures implement Stochastic Gradient Descent (SGD) with momentum  $\mu = 0.9$  and learning rate  $lr = 0.01$ .

All the results of the tests are shown in Table 3, where we can notice that SpecNet outperforms both its 2-channels version and the naïve classifier (which is referred to as “LSTM” in the table). The displayed values are the average accuracies obtained in cross validation using 5 folds.

The same table reports the results of an additional experiment in which the models are pre-trained on all the available subjects in ActionSense for 100 epochs, and then fine-tuned on S04 for 100 additional epochs. We can notice that the

| Model      | Pre-trained | Epochs | Accuracy     |
|------------|-------------|--------|--------------|
| LSTM       | -           | 100    | 76.19        |
| SpecNet2Ch | -           | 100    | 72.14        |
| SpecNet    | -           | 100    | <b>85.48</b> |
| LSTM       | ✓           | 100    | 42.86        |
| SpecNet2Ch | ✓           | 100    | 73.33        |
| SpecNet    | ✓           | 100    | <b>90.24</b> |

Table 3. Comparison of the three proposed models that classify the EMG signals with and without pre-training.

spectrogram networks achieve a non-negligible boost in accuracy, while the accuracy of the naïve LSTM model drastically drops; we suppose that this is due to the high differences in the raw signals coming from different subjects.

**Multimodal Architecture.** Finally, we combine the best performing networks presented in the previous sections to realize a multimodal architecture: MultiNet. We choose to take the output of the last average pooling layer of SpecNet to represent the EMG data, and to use the concatenation as the technique to merge together both the RGB features coming from different clips and the features related to different modalities. The detailed architecture is represented in Figure 5.

The performances of Multinet are analyzed in the next section.

## 4.2. Results

Table 5 contains the results of all our experiments regarding MultiNet, during which we have tried both to tune the parameters affecting the training process (optimizer and

| Input                    | Operator       | BN | NL | $s$ |
|--------------------------|----------------|----|----|-----|
| $16 \times 17 \times 94$ | conv2d, 3x3    | ✓  | HS | 1   |
| $16 \times 8 \times 47$  | maxpool2d, 2x2 | -  | -  | 2   |
| $16 \times 8 \times 47$  | conv2d, 3x3    | -  | HS | 1   |
| $16 \times 8 \times 47$  | avgpool2d, 2x2 | -  | -  | 2   |
| $16 \times 4 \times 23$  | conv2d, 3x3    | -  | HS | 1   |
| $16 \times 4 \times 23$  | conv2d, 3x3    | ✓  | HS | 1   |
| $32 \times 4 \times 23$  | avgpool2d, 2x2 | -  | -  | 2   |
| $32 \times 2 \times 11$  | flatten        | -  | -  | -   |
| $1 \times 1 \times 704$  | linear         | -  | HS | -   |
| $1 \times 1 \times 512$  | linear         | -  | -  | -   |
| $1 \times 1 \times 21$   | softmax        | -  | -  | -   |

Table 4. Specification of SpecNet. The input is in the form (C, H, W) where C is the number of channels, H the height, W the width. BN denotes batch normalization, NL denotes the type of nonlinearity used, HS stands for hardswish,  $s$  denotes the stride.

learning rate) and to make a comparison with the single modalities taken by themselves. The highest accuracy (90.48%) is reached by the multimodal classifier when exploiting the pre-training of SpecNet with the implementation of the hardswish and using SGD as optimizer with parameters  $\mu = 0.9$  and  $lr = 0.01$ .

In general, it can be noticed that the accuracy of the multimodal classifier is only slightly higher than the one achieved by SpecNet; this can be due to the fact that there is a huge gap between the performances of the models using only EMG data and the ones using only RGB data (max accuracy for RGB is 64.05%, max accuracy for EMG is 90.24%), and that on average the norm of EMG features

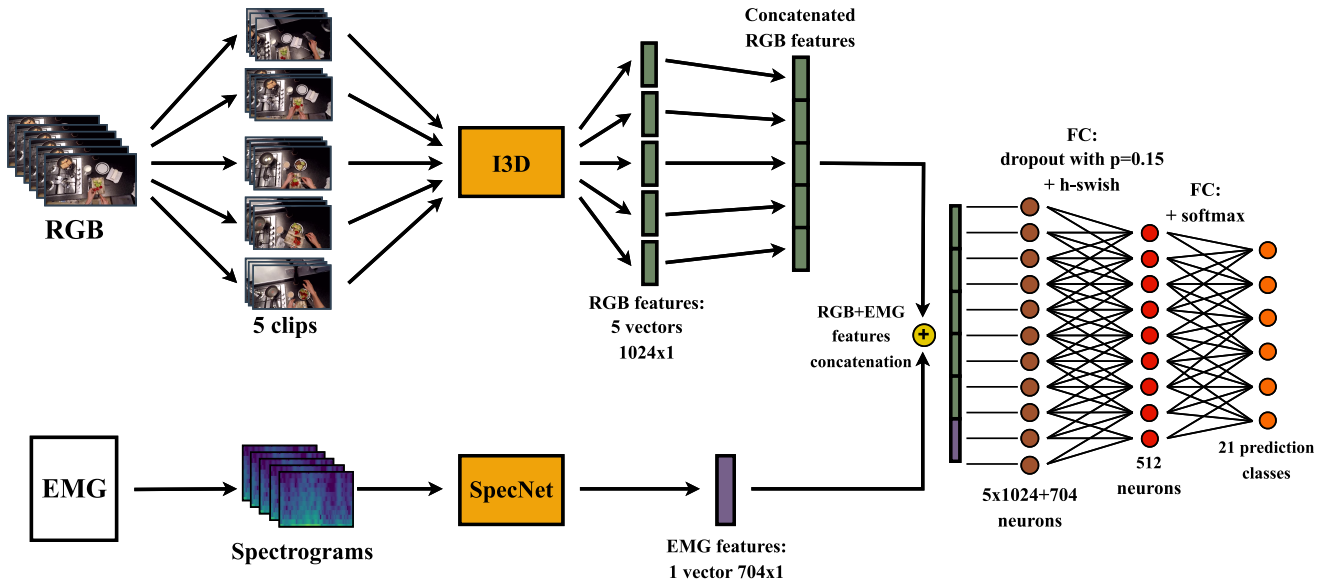


Figure 5. Our proposal for the final multimodal architecture, MultiNet.

| Model          | Modality | Optimizer | Parameters              | Epochs | Accuracy     |
|----------------|----------|-----------|-------------------------|--------|--------------|
| MultiNet       | RGB      | SGD       | $\mu = 0.9, lr = 0.01$  | 100    | 64.05        |
| MultiNet       | EMG      | SGD       | $\mu = 0.9, lr = 0.01$  | 100    | 85.48        |
| MultiNet       | RGB+EMG  | SGD       | $\mu = 0.9, lr = 0.01$  | 100    | 81.67        |
| MultiNet+PT+HS | RGB+EMG  | SGD       | $\mu = 0.9, lr = 0.001$ | 100    | 86.67        |
| MultiNet+PT+HS | RGB+EMG  | Adam      | $lr = 0.001$            | 100    | 88.33        |
| MultiNet+PT+HS | RGB+EMG  | SGD       | $\mu = 0.9, lr = 0.005$ | 100    | 88.57        |
| MultiNet+PT    | RGB+EMG  | SGD       | $\mu = 0.9, lr = 0.01$  | 100    | 89.28        |
| MultiNet+PT+HS | EMG      | SGD       | $\mu = 0.9, lr = 0.01$  | 100    | 90.24        |
| MultiNet+PT+HS | RGB+EMG  | SGD       | $\mu = 0.9, lr = 0.01$  | 100    | <b>90.48</b> |

Table 5. MultiNet experimental results for both unimodal and multimodal analysis. PT stands for Pre-Trained when fine-tuning is applied on the spectrogram part. HS stands for hardswish where this activation function is implemented in the place of ReLU. The results are computed by taking the average of the 5 folds cross validation accuracies.

is higher than the one of RGB features (making them affect more the predictions).

Finally, we think that the accuracies obtained on ActionSense in general are so high because of the reduced dimensions of the dataset.

## 5. Conclusion

We performed a study on a first basic approach at multi-modal egocentric action recognition, and explored the possibility of processing data coming from electromyographic wearable sensors through 2D CNNs by computing their spectrograms. In our analysis we could observe the high efficiency of a relatively shallow network like SpecNet when performing a classification task on a small dataset, and that it can be beneficial when used in combination with other features in a multimodal architecture.

Our work could be improved by conducting further analyses on:

- Other modalities, like optical flow, audio, tactile data, eye tracking or head motion.
- Other architectures such as residual networks (ResNets) or region-based CNNs (R-CNN).
- Different and/or bigger datasets. For instance, one can consider to exploit all the available data from ActionSense, like we did for the pre-training of SpecNet.

## References

- [1] Muhammad Abdullah, Mobeen Ahmad, and Dongil Han. Facial expression recognition in videos: An cnn-lstm based model for video classification. In *2020 International Conference on Electronics, Information, and Communication (ICEIC)*, pages 1–3. IEEE, 2020. 1
- [2] Md Al-Amin, Wenjin Tao, David Doell, Ravon Lingard, Zhaozheng Yin, Ming C Leu, and Ruwen Qin. Action recognition in manufacturing assembly using multimodal sensor fusion. *Procedia Manufacturing*, 39:158–167, 2019. 2
- [3] Joao Carreira and Andrew Zisserman. Quo vadis, action recognition? a new model and the kinetics dataset. In *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 6299–6308, 2017. 1, 2, 4
- [4] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, and Michael Wray. Scaling egocentric vision: The epic-kitchens dataset. In *European Conference on Computer Vision (ECCV)*, 2018. 1
- [5] Alexandre De Brebisson and Pascal Vincent. An exploration of softmax alternatives belonging to the spherical loss family. arXiv preprint arXiv:1511.05042, 2015. 4
- [6] Joseph DelPreto, Chao Liu, Yiyue Luo, Michael Foshey, Yunzhu Li, Antonio Torralba, Wojciech Matusik, and Daniela Rus. Actionsense: A multimodal dataset and recording framework for human activities using wearable sensors in a kitchen environment. In S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems*, volume 35, pages 13800–13813. Curran Associates, Inc., 2022. 1, 3, 5
- [7] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009. 2
- [8] Guillermo Garcia-Hernando, Shanxin Yuan, Seungryul Baek, and Tae-Kyun Kim. First-person hand action benchmark with rgb-d videos and 3d hand pose annotations. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 409–419, 2018. 2
- [9] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, et al. Searching for mobilenetv3. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1314–1324, 2019. 5
- [10] Evangelos Kazakos, Arsha Nagrani, Andrew Zisserman, and Dima Damen. Epic-fusion: Audio-visual temporal binding for egocentric action recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5492–5501, 2019. 2, 4

- [11] Zhenyang Li, Kirill Gavriluk, Efstratios Gavves, Mihir Jain, and Cees GM Snoek. Videolstm convolves, attends and flows for action recognition. *Computer Vision and Image Understanding*, 166:41–50, 2018. [2](#)
- [12] Ji Lin, Chuang Gan, and Song Han. Tsm: Temporal shift module for efficient video understanding. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7083–7093, 2019. [1](#)
- [13] Irene Mendez, Bo Winther Hansen, Christine Marie Grabow, Erik Joannes Leivsgard Smedegaard, Nanna Bloch Skogberg, Xenia Jørgensen Uth, Astrid Bruhn, Bo Geng, and Ernest Nlandu Kamavuako. Evaluation of the myo arm-band for the classification of hand motions. In *2017 International Conference on Rehabilitation Robotics (ICORR)*, pages 1211–1214. IEEE, 2017. [1](#)
- [14] Jonathan Munro and Dima Damen. Multi-modal domain adaptation for fine-grained action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 122–132, 2020. [1](#)
- [15] Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3m: A universal visual representation for robot manipulation. *arXiv preprint arXiv:2203.12601*, 2022. [1](#)
- [16] Suriya Singh, Chetan Arora, and CV Jawahar. Trajectory aligned features for first person action recognition. *Pattern Recognition*, 62:45–55, 2017. [1](#)
- [17] Sibo Song, Vijay Chandrasekhar, Ngai-Man Cheung, Sanath Narayan, Liyuan Li, and Joo-Hwee Lim. Activity recognition in egocentric life-logging videos. In *Computer Vision-ACCV 2014 Workshops: Singapore, Singapore, November 1-2, 2014, Revised Selected Papers, Part III 12*, pages 445–458. Springer, 2015. [1](#)
- [18] Vignesh Thakkar, Suman Tewary, and Chandan Chakraborty. Batch normalization in convolutional neural networks — a comparative study with cifar-10 data. In *2018 Fifth International Conference on Emerging Applications of Information Technology (EAIT)*, pages 1–5, 2018. [5](#)
- [19] Ryan J Visee, Jirapat Likitlersuang, and Jose Zariffa. An effective and efficient method for detecting hands in egocentric videos for rehabilitation applications. volume 28, pages 748–755. IEEE, 2020. [1](#)
- [20] Xinming Wang, Weili Ding, Shuangshuang Bian, and Honghai Liu. Action recognition through fusion of semg and skeletal data in feature level. *Journal of Ambient Intelligence and Humanized Computing*, 13(9):4125–4134, 2022. [2](#)